

2(a)(i)	work done per unit mass	B1
	work (done) moving mass from infinity (to the point)	B1
2(a)(ii)	$\phi = -GM/r$ $= - (6.67 \times 10^{-11} \times 7.3 \times 10^{22}) / (1.7 \times 10^6)$	C1
	$= - 2.9 \times 10^6 \text{ J kg}^{-1}$	A1
2(b)(i)	$E_P = m\phi$	B1
2(b)(ii)	$\frac{1}{2}mv^2 + m\phi = 0$	M1
	correct algebra leading to $v = \sqrt{-2\phi}$	A1
2(c)	speed $= \sqrt{2 \times 2.9 \times 10^6}$ $= 2400 \text{ m s}^{-1}$	A1
2(d)	$\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$	C1
	$3.34 \times 10^{-27} \times \langle c^2 \rangle = 3 \times 1.38 \times 10^{-23} \times 400$	C1
	$c_{\text{r.m.s.}} = 2200 \text{ m s}^{-1}$	A1
2(e)	r.m.s. speed is an average so many molecules have speeds greater than the escape speed or there is a distribution of molecular speeds (around the r.m.s. value) so many molecules have speeds greater than the escape speed	B1